

Element C - Presentation and Justification of Solution Design Requirements

Project Title: MJ's Last Stand (Echo Rack)

Design Requirement #1 (HIGH Priority)

- The highest priority would be that the stand is able to hold the weight of our gong. This is after all the entire point of our project. If the stand can not hold the weight of the gong then it would break and our whole project would be for nothing.
- Prior research in types of wood could be helpful in determining what kind of wood would be able to hold the most weight while being relatively inexpensive. Our structure research would also be helpful in determining the best design to hold the gong effectively. Our current design isn't terrible and we might be able to reuse it with a different material (being wood).
- Mr. Waller doesn't seem to care what it looks like as long as it isn't ugly. I think he mostly cares about the functionality of the gong. He wouldn't care about the wood type because we'd likely be painting over it anyway. We will show him a design or sketch and see if he has any feedback on how the structure might clash with aesthetic appeal.

Gong needs to hung & be Balanced (HIGH Priority) #2

- What it needs to do: The gong stand needs to be suspended hanging in the air from the metal/rubber straps connected to the top bar, it also needs to be balanced hanging, as in not slanted to either side, but perfectly suspended in the middle.
- Why it's important: The angle of where the gong is hit causes different sounds to come out, so it needs to be balanced in the air, so that when it is hit, it releases a the sound intended by the user, which it won't do once if it's slanted.
- Stakeholder input: Mr Waller shared concerns about the way that the gong was slanted, fearing that there could be a slip up during a concert if it is hit the wrong way, this would effectively solve this issue.

Design Requirement #3 Elegant design (HIGH Priority)

- **What is needs to do:** The gong stand needs to look elegant and special. They ERHS band needs a gong stand that is different from other ordinary stands.
- **Why its important:** This is important becuae the gong stand needs to be colorful and different. The ERHS band needs an elegant design that stands out and is different.
- **Stakeholder Input:** Mr Waller shared that he wants an elegant gong stand for the ERHS band. The gong stand needs to be colorful and stand out toward the audience.

Design Requirement #4 (HIGH Priority)

- Additionally, another high priority requirement would be an integrated mute into the gong stand. Although the gong stand does not need a muting feature, it might be helpful to have when performing so you don't have to hug the gong to make it stop ringing. It looks unprofessional. The ideal solution would be to have a foot pedal that control felt clasps latching on and off to the gong to prevent it from ringing.
- Based on our research, the design that we were wanting to build shouldn't be too difficult seeing as many rough percussion instrument mutes use very similar systems. Of course they aren't foolproof and don't always mute it thoroughly (this is because there still needs to be a place to strike the gong, so the felt can't take up the entire space), something that at least mutes it enough will suffice for our needs.
- While at first I thought we should keep the mute as a surprise for Mr. Waller, I'm starting to think now maybe we shouldn't. (we'll talk to him about that) I'd be willing to bet money that he'd be on board with the idea for numerous reasons but just to check we'll ask again. Mr. Waller is big on professionalism, so having a sleek, non-interruptive way to mute the gong would be something that he is looking for.

Design Requirement #5 (HIGH Priority)

- The design of the gong stand must be low cost; not only for the sake of making the project not too expensive, but also make an accessible product.
- Throughout our research, it was realized that gong stands are usually very expensive ranging from hundreds of dollars to thousands. Producing a low-cost, yet effective product of a gong stand will appeal to the market.
- In our interview with Mr. Waller, he understood that we did not have access to unlimited money or material. For that reason, it was concluded that a low-cost design would be best considering our capabilities.

Design Requirement #6 (LOW Priority)

- The literal deadline for the gong stand by the April concert; the gong stand should be built and tested beforehand.
- This does not pertain to research, as it is simply the nature of the gong stand that it must be used for performances. The gong instrument will be used several times throughout the April concert, so the stand should be completed by then.
- Mr. Waller confirms that he wishes for the stand to be completed by the concert. Of course, if we do not complete the stand, it may still be used for future years and perhaps by the orchestra or other performing groups.

Environmentally Friendly and Disposable (Low Priority) #7

- What it needs to do: The gong stand needs to be environmentally friendly and easily disposable. Our invention, should something go wrong should be easily disposable, and when disposed off it should decompose in the environment or landfill that it is thrown in.

- Why it's important: There was another band invention made out of wood that soon grew mold after years of use and it was too difficult to remove so it is currently sitting in the closet of the band room, our invention should be easily disposable, and hopefully wouldn't just sit in a shed waiting to be disposed of.
- Stakeholder input: Mr Waller shared concerns about the gong stand growing mold, and if it did grow mold, how would they dispose of it because of it's weight, and also how would it fair in different environmental conditions..